Secure wiring and finish the design

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Document permanent connections and evaluate added mass or friction.

SUCCESS CRITERIA - I can...

1. Check circuit continuity and mechanical security.
2. Preserve moving-part clearance during decoration.
3. Relate a design revision to an energy transfer or force.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “strain relief” and “insulation” to explain your model or result. Sentence starters are optional.

strain relief: support that keeps a pull from stressing an electrical joint

insulation: material that limits unwanted electric conduction

short circuit: an unintended low-resistance path bypassing a component

01 Warm-up: bring back an earlier idea

At unchanged wheel rpm, circumference becomes 4 times as large.
Compare ideal travel speed and state one real-world limitation.

Word help: circumference = distance around the edge of a circle

Sentence starter: Both … . They differ in … because … .

Marzano check - circle: ① ② ③

02 Worked example: explain the model

Reference: I = 0 with power disconnected.
Disconnect before soldering/securing wires. Use insulation and strain relief; route wires clear of gears. Follow supervised tool procedures.

Worked reasoning: A secure circuit also needs strain relief: a pull on a wire should not be carried solely by a solder joint. Insulation prevents exposed conductors from making unintended contacts.

Your turn: Draw one wire route showing a connection, insulation, strain relief and clearance from a rotating axle.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

03 Read and interpret the model

Draw your final wiring route.
Mark soldered joints, insulation, strain relief and the switch location.
Identify one possible wheel or gear interference point.

Sentence starter: The … in my model represents … . Its label shows … .



Marzano check - circle: ① ② ③

04 Apply the physics

A model car’s mass increases from 200 g to 250 g after decorations are added.
Calculate added mass and fill in the masses in kilograms.
Predict how acceleration changes if net forward force stays the same.
Explain why that assumption needs checking.

Sentence starter: I predict … . My model supports this because … .

Design	Mass (kg)	Net force (N)	Acceleration
Before	____	____	____
After	____	____	____

Marzano check - circle: ① ② ③

05 Explain your reasoning

A wire rubs a wheel.
Explain the possible effect on energy transfer and propose a repair before another powered test.

Sentence starter: The step I would change is … . The evidence is … .

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

06 Review: connect an earlier idea

At fixed current, move from r to 4 r from a long straight wire. Using B proportional to 1/r, compare field strength. **Mark** the distances and field directions you compare.

Word help: current = rate of charge flow

Sentence starter: Both … . They differ in … because … .

Current out of page (dot). Add field arrows.



Marzano check - circle: 1 2 3

07 Exit check

Record one quality check your partner completed and one revision supported by that check.

Sentence starter: I notice … . This supports … because … .



Marzano check - circle: 1 2 3

08 STRETCH: test the model

Create a power-off inspection checklist with three specific observations that would make you stop before powering the car.
Explain the concern behind each.

Sentence starter: My claim is … . The diagram/data show …, so … .



Marzano check - circle: 1 2 3